The Project Title

A Study on Speech Emotion Recognition Based on Multi-Feature and Deep Models

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Abstract

This dissertation addresses the critical challenge of robust Speech Emotion Recognition (SER) in the presence of environmental noise. While deep learning has significantly advanced SER, model performance often degrades when moving from clean, laboratory-recorded data to more realistic, noisy conditions. This study conducts a systematic evaluation of deep learning models and audio features to improve noise robustness and generalization. We investigate four distinct architectures: a Multilayer Perceptron (MLP), a shallow Convolutional Neural Network (CNN), a deeper ResNet-18, and the attention-based Audio Spectrogram Transformer (AST). These models are trained on both traditional Mel-frequency cepstral coefficient (MFCC) features and 2D Mel-spectrograms.

The core of our methodology revolves around a comprehensive on-the-fly data augmentation strategy, where training data is mixed with white noise and natural soundscapes from the ESC-50 dataset at various signal-to-noise ratios (SNRs). We evaluate model performance on two standard benchmarks, RAVDESS and IEMOCAP, under both clean and noisy test conditions. The findings consistently demonstrate that the AST architecture surpasses the convolutional and MLP-based models, particularly when trained with data augmentation. Augmentation not only preserves but often improves AST's performance on clean test sets while significantly enhancing its robustness against a wide range of noises and SNRs.

Furthermore, our analysis reveals a crucial interaction between model architecture and augmentation strategy. While AST effectively leverages diverse acoustic perturbations to learn noise-invariant features, the ResNet-18 model proves more fragile, exhibiting performance degradation under the same conditions. The study also highlights dataset-specific effects, where augmentation leads to a more balanced precision-recall trade-off on the heterogeneous IEMOCAP dataset, particularly for the AST model. Ultimately, this research underscores the synergy between transformer architectures and robust data augmentation, presenting a promising framework for developing more practical and generalizable SER systems.

Acknowledgements

Abbreviations

AST	Audio Spectrogram Transformer
CCC	concordance correlation coefficient
CNN	convolutional neural network
DCT	discrete cosine transform
DFT	discrete Fourier transform
DTs	Decision Trees
FFN	Feed-forward network
GMMs	Gaussian Mixture Models
HMMs	Hidden Markov Models
HSFs	high-level functional statistics
KNN	K-nearest neighbors
LLDs	low-level descriptors
MFCCs	Mel-frequency cepstral coefficients
MHSA	multi-head self-attention
MLP	multilayer perceptron
RMS	root-mean-square
SER	Speech Emotion Recognition
SNR	signal-to-noise ratio
SVMs	Support Vector Machines
TEO	Teager Energy Operator
UAR	unweighted average recall
ViT	Vision Transformer

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CHAPTER 1

Introduction

1.1 Background

Speech conveys rich affective information through prosody, articulation, and spectral characteristics, enabling listeners to infer speakers' intentions and internal states. Speech Emotion Recognition (SER) aims to computationally infer human emotion from acoustic signals, either as discrete categories (e.g., anger, happiness, sadness, fear, and neutral) or along continuous dimensions (e.g., valence, arousal, and dominance) [11] . Rooted in speech processing, affective computing, and psychology, early SER systems relied on handcrafted prosodic and spectral features combined with classical classifiers [15]. Over the past decade, deep learning has transformed the field: convolutional networks on time–frequency representations, recurrent and attention-based models for temporal dynamics, and, more recently, transformer architectures and self-supervised pretraining have markedly improved generalization [15, 43, 39, 13, 3]. The research focus has also shifted from acted laboratory corpora to spontaneous, in-the-wild data, emphasizing robustness to noise, channel variability, and cultural–linguistic diversity. In parallel, there is growing attention to transparency, privacy, and fairness, reflecting the societal implications of emotion inference technologies [14].

SER has broad practical value. In customer service and contact centers, emotion-aware analytics can monitor user satisfaction and support agent coaching in real time [35]. In healthcare and mental health, non-invasive acoustic markers assist in screening and longitudinal monitoring of affective disorders, stress, and burnout. In human–computer interaction, conversational assistants and social robots can adapt responses and dialogue strategies based on users' emotional cues [12]. In automotive and smart environments, continuous monitoring of driver frustration or fatigue enhances safety and personalization. Education technology, entertainment, and recommendation systems also benefit from affect-aware feedback and content curation [40]. These applications motivate solutions that are accurate, low-latency, and resource-efficient for both cloud and on-device deployment.

The research significance of SER is twofold. Scientifically, SER advances our understanding of how affect is encoded in speech and how to learn robust, interpretable representations under limited, imbalanced, and noisy supervision [19, 20]. Technically, it drives innovation in domain generalization, cross-lingual transfer, and multimodal fusion with text and vision, while promoting explainability to build user trust [1]. Addressing ethical and legal considerations—including data privacy, informed consent, and bias mitigation—is essential to responsible deployment. Finally, rigorous benchmarking, reproducibility, and standardized evaluation protocols (e.g., unweighted/weighted accuracy for categorical tasks and concordance correlation for continuous estimation) are critical for cumulative progress. Altogether, SER research provides both theoretical insights and practical tools for empathetic, human-centered intelligent systems.

1.2 Main Contents of the Dissertation

This dissertation addresses the challenge of suboptimal recognition accuracy in Speech Emotion Recognition by developing a deep learning based framework that integrates speech signal preprocessing, feature engineering, attention mechanisms, and model optimization. We conduct a systematic study of SER methods from multiple perspectives, including audio preprocessing, data augmentation, feature selection, model architectures, and evaluation metrics. The central objective is to learn feature representations with stronger affective discriminability directly from speech, thereby improving accuracy, robustness, and generalization across speakers, recording conditions, and datasets. The proposed approach emphasizes practical deployability by balancing performance with computational efficiency for both cloud and on-device inference.

1.3 Structure of the Dissertation

The dissertation is organized as follows. Chapter 1 introduces the research background and motivation, identifies the key challenges in SER, and outlines the scope and structure of the thesis. Chapter 2 surveys related work on speech signal processing, feature extraction, the design of attention mechanisms, and emotion taxonomies and recognition paradigms, summarizing major international developments. Chapter 3 presents the methodology. We construct features from audio data using Mel-frequency cepstral coefficients (MFCCs) and Mel-spectrograms; we then investigate multiple models for SER, including a multilayer perceptron (MLP), a shallow convolutional neural network (CNN), ResNet-18 for 2D Mel-spectrogram inputs, and the Audio Spectrogram Transformer (AST), a ViT-based model for audio classification. Finally, we apply data augmentation by injecting noise during training to improve generalization and mitigate overfitting. Chapter 4 describes the experiments. We design comparative studies to evaluate the performance differences among features and models on SER tasks under consistent protocols and metrics. Chapter 5 provides discussion. We analyze the experimental results in depth, examine the relative strengths and weaknesses of different features and models, and articulate limitations and directions for future research. Chapter 6 concludes the thesis by summarizing the main findings and contributions, and highlighting potential avenues for subsequent work.

CHAPTER 2

Literature Review

2.1 Datasets for Speech Emotion Recognition

Datasets are foundational to empirical research in SER. They can be broadly categorized into acted corpora and spontaneous or in-the-wild corpora. Acted datasets are typically recorded in controlled laboratory environments with professional actors performing target emotions. While these corpora offer clean signals and clearer affective cues that facilitate modeling, they may over-represent exaggerated expressions and fail to capture the subtleties of everyday speech. By contrast, spontaneous corpora contain naturalistic affect produced in real interactions or semi-scripted dialogues. They better reflect practical deployment scenarios but introduce additional challenges such as background noise, speaker overlap, label ambiguity, and class imbalance. Below we summarize several widely used SER datasets.

RAVDESS. The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) [28] contains recordings from 24 professional actors (12 female, 12 male) producing two lexically matched statements across eight emotions (neutral, calm, happy, sad, angry, fearful, disgust, surprise) with normal and strong intensities (neutral has no intensity). The speech subset totals 1,440 audio files, recorded with consistent studio conditions, and is commonly used for speaker-independent evaluation. RAVDESS provides balanced classes and clear prosodic cues, making it suitable for benchmarking and ablation studies; however, its acted nature can limit ecological validity when models are deployed on spontaneous speech.

IEMOCAP. Collected at the University of Southern California, IEMOCAP [8] is a multimodal corpus containing speech, video, facial motion capture, and transcripts over approximately 12 hours of dyadic interactions with 10 actors (five female, five male) arranged into five sessions (each session features one female–male pair). The corpus includes both scripted and improvised dialogues. Utterances are annotated by multiple raters for nine discrete emotions (anger, happiness, excitement, sadness, frustration, fear, surprise, neutral, and other) as well as three continuous dimensions (valence, arousal, dominance). In practice, many studies merge excitement into happiness, yielding a four-class set (neutral, angry, happy, sad) or a six-class set (neutral, angry, happy, sad, frustration, surprise). For comparability, this thesis adopts both the four-class and six-class protocols in line with prior work.

EmoDB. The Berlin Database of Emotional Speech [7] comprises recordings from 10 German actors (five female, five male) who produced 10 phonetically balanced sentences with seven target emotions: anger, sadness, happiness, fear, neutral, disgust, and boredom. After rigorous quality screening, around 500 utterances were retained. Audio was recorded in a studio at 48 kHz, 16-bit, and is often downsampled to 16 kHz for processing. Although acted, the dataset emphasizes natural-sounding realizations through actor induction techniques, providing strong, well-separated emotional prototypes.

MELD. The Multimodal EmotionLines Dataset [32] extends EmotionLines by adding audio and video modalities to textual dialogues. Derived from the TV show "Friends," MELD contains over 1,400 multi-speaker conversations and more than 13,000 utterances, annotated with seven emotions: anger, disgust, sadness, joy, neutral, surprise, and fear. Its multimodal, conversational structure enables the study of contextual emotion dynamics and speaker interactions beyond isolated utterances.

2.2 SER with Acoustic Features

Acoustic feature engineering plays a central role in early SER pipelines. It compresses raw waveforms into compact descriptors derived from time-domain and frequency-domain analyses, guided by speech science. Canonical categories include prosodic features (pitch, energy, duration), voice-quality features (jitter, shimmer, harmonicity), spectral features (formants, cepstral coefficients), and Teager Energy Operator (TEO) based features. These low-level descriptors (LLDs) capture frame-wise micro-variations in speech and can be summarized into high-level functional statistics (HSFs) such as means, first- and second-order derivatives, and other moments to encode utterance-level context.

Empirical studies have shown that acoustic features carry substantial affective information, reflecting changes in emotion, speaker identity, speaking style, and tempo [27]. Lee et al. [21] and Schuller et al. [41] reported strong results using feature sets combining energy-, pitch-, and spectral-related cues, with pitch-related features often proving more discriminative than energy features. Rao et al. [34] compared LLDs against HSFs (and their combination) and found that combining frame-level prosody (e.g., F0 trajectories, energy) with statistical summaries improves recognition performance. Lugger et al. integrated prosodic and voice-quality features (e.g., glottal gradient, skewness gradient, closure rate gradient) within a two-stage classifier to reach 74.5% accuracy. Li et al. [24] further augmented HSFs with jitter and shimmer to boost performance. Zhang et al. [50] combined prosodic with voice-quality features (including the first three formants), improving accuracy by about 10% over prosody-only baselines. Sato et al. [38] used clustering to label frames and constructed a segmented MFCC feature set, outperforming K-nearest neighbors (KNN) on prosody and Hidden Markov Models (HMMs) on conventional MFCCs. Bitouk et al. [6] introduced new spectral features by computing MFCCs separately for stressed vowels, unstressed vowels, and consonants, concluding that consonants may carry richer affective cues than vowels. Zhou et al. [51] proposed three TEO variants—especially a standard-band TEO autocorrelation envelope—to study airflow energy changes under stress, achieving better results than pitch+MFCC combinations.

After feature extraction, classifiers are typically used for final affect prediction. Traditional approaches include Gaussian Mixture Models (GMMs), Support Vector Machines (SVMs), and Decision Trees (DTs) . More recent methods employ deep learning to learn discriminative affect representations from acoustic features, such as 1D CNNs and RNNs, trained with task-appropriate loss functions.

Overall, acoustic-feature-based SER offers two key advantages. First, it provides a degree of interpretability: models can assign higher weights to features with stronger affective discriminability, which often aids accuracy compared to purely black-box baselines. Second, system complexity can be relatively low: feature extraction is algorithmic with few learnable parameters, while deep architectures built on these features improve adaptability and reduce overfitting, enabling transfer across related affective tasks. Despite progress, several issues remain: (1) many features rely on the short-time Fourier transform (STFT), a global transform that may inadequately represent the inherently non-stationary, localized time–frequency patterns of affect; (2) attention mechanisms operating on handcrafted features may struggle to highlight truly salient regions, potentially discarding relevant emotional cues.

2.3 SER with Spectrogram-based Features

A central question in spectrogram-based SER is how to extract affectively discriminative representations from time–frequency images. Mel-spectrograms differ from classical handcrafted features (e.g., formants, jitter, MFCCs) in that they preserve localized time–frequency structure and delegate hierarchical feature learning to deep networks. Representing energy over time and frequency in a 2D grid enables learning of complex spectro-temporal patterns that are difficult to encode with fixed handcrafted descriptors [31]. During training, networks can discover filter-like patterns akin to Mel filterbanks while also capturing long-range dependencies and contextual cues typically missed by shallow features.

Since the 2012 deep learning breakthroughs on ImageNet, advances in model capacity and data processing have translated to SER. Compared with conventional machine learning, deep models can learn affect representations without manual feature selection. Feed-forward deep neural networks (DNNs) comprise multiple hidden layers whose weights are optimized via gradient descent and backpropagation to minimize a task loss. Convolutional neural networks (CNNs) are particularly impactful in SER , as spectrograms exhibit strong local correlations along time and frequency; local receptive fields allow CNNs to extract progressively higher-level features from 2D inputs [33]. Recurrent neural networks (RNNs) and Long Short-Term Memory (LSTM) networks are commonly stacked on top of CNN features to model temporal dependencies and can also operate directly on frame-level acoustic features to yield high-level affect representations [4]. Attention mechanisms [44] further refine these representations by emphasizing salient regions conducive to emotion discrimination.

Rapid progress has been reported in recent years . Trigeorgis et al. [43] and Lim et al. [26] proposed CNN–LSTM hybrids in which CNNs extract high-level spectrogram features and LSTMs capture sequence dynamics, outperforming traditional machine learning; similar architectures have been widely adopted [37, 43]. To model spatial relationships in spectra, capsule networks based on CNNs have also been explored , achieving promising sentence-level affect modeling [45]. Representative spectrogram-based deep architectures for SER are summarized in 2.1.

Table 2.1: Related research on SER using spectrogram-based models

Study	Year	Method	Dataset	Result
Zou et al. [9]	2014	DBN	ABC Corpus	Accuracy: 52.2%
Kim et al. [16]	2019	DNN	IEMOCAP	UAR: 61.4%
Aldeneh et al. [2]	2017	CNN	IEMOCAP; MSP-IMPROV	UAR: 52.6%
Lee et al. [22]	2015	BiLSTM	IEMOCAP	UAR: 63.89%
Latif et al. [18]	2019	CNN-LSTM	IEMOCAP; MSP-IMPROV	UAR: 60.23%; 52.43%
Bhosale et al. [5]	2020	ACNN	IEMOCAP	UAR: 63.15%
Mustaqeem et al. [36]	2020	CNN-LSTM	EMODB	Accuracy: 85.57%

On top of high-capacity encoders, attention mechanisms have been extensively used to further improve accuracy , including self-attention [44], local attention [29], hop-wise attention [49], and many other variants [30, 48]. These modules enhance weights of informative time–frequency regions while suppressing irrelevant ones, thereby focusing the network on discriminative patterns; see Table 2.2 for recent studies leveraging attention to boost SER.

Table 2.2: Related research on SER using attention-based mechanisms

Study	Year	Method	Dataset	Result
Mirsamadi et al. [29]	2017	Local Attention	IEMOCAP	UAR: 58.8%
Li et al. [25]	2019	Self Attention	IEMOCAP	UAR: 58.1%
Yoon et al. [49]	2019	Multi-Hop Attention	IEMOCAP	UAR: 65.2%
Xie et al. [46]	2019	TF Attention	CASIA; eNTERFACE	UAR: 63.89%; 89.6%
Tarantino et al. [42]	2019	Self Attention	IEMOCAP	UAR: 63.8%
Xu et al. [47]	2020	Alignment Attention	IEMOCAP	UAR: 57.4%
Li et al. [23]	2021	Self BiAttention	EMODB	Accuracy: 85.95%

Finally, learned affect representations are mapped to task outputs via suitable loss functions and evaluation metrics. For categorical SER, cross-entropy remains the dominant training objective, while mean squared error is often used for dimensional emotion regression. Historically, accuracy was the default metric; however, as datasets grew and class imbalance became prevalent, the Interspeech 2009 Emotion Challenge popularized unweighted average recall (UAR), computed as the mean of per-class recalls, which is now widely adopted for imbalanced settings. For dimensional SER, the concordance correlation coefficient (CCC) is commonly used. Steady gains from stronger encoders and attention, yet several challenges remain: (1) fixed-size convolutional kernels can restrict the effective receptive field over salient emotional regions, limiting global context capture; (2) attention layers may over-rely on fully connected heads and softmax, under-exploiting local, global, and positional cues; (3) cross-entropy is sensitive to label distributions and can struggle when class manifolds are close; (4) variable-length utterances complicate segment-level training and can misalign segment labels with utterance-level annotations.

CHAPTER 3

Methodology

3.1 Feature Extraction

Feature extraction plays a crucial role in speech emotion recognition. A variety of acoustic features can be derived from emotional speech to reflect characteristics of a speaker's affective behavior. High-quality and informative features have a significant impact on recognition accuracy. How to extract and select features that effectively capture emotional variation remains one of the most important problems in the current field of speech emotion recognition.

This section details the feature extraction pipeline and unified notation used throughout the thesis. Let $x[n]$ denote the discrete-time speech signal sampled at rate f_s (Hz). Frames are indexed by m, frequency bins by k, and Mel bands by f. The frame length is N samples, hop size is H samples, the analysis window is $w[n]$ (Hann unless otherwise stated), the discrete Fourier transform (DFT) length is K, and the number of Mel filters is F. We use $X[k, m]$ for the short-time spectrum, $P[k, m] = |X[k, m]|^2$ for the power spectrum, $\mathbf{B} \in \mathbb{R}^{F \times K}$ for the Mel filterbank, and a small constant $\varepsilon > 0$ for numerical stability.

3.1.1 MFCC

Mel-frequency cepstral coefficients (MFCCs) [10] are compact features that approximate human auditory perception by (i) warping frequency to the Mel scale, (ii) applying a band-pass filterbank to obtain perceptual energies, (iii) compressing the dynamic range, and (iv) decorrelating energies via a discrete cosine transform (DCT). The standard pipeline is as follows.

1) Pre-emphasis (optional) enhances high-frequency components:

$$y[n] = x[n] - \alpha\, x[n-1], \quad \alpha \in [0.95, 0.99]. \tag{3.1}$$

2) Framing and windowing create approximately stationary short-time seg-

ments:

$$X[k,m] = \sum_{n=0}^{N-1} x[n+mH]\, w[n]\, e^{-j2\pi kn/K}, \quad 0 \le k < K. \tag{3.2}$$

3) Power spectrum is computed as

$$P[k,m] = \frac{|X[k,m]|^2}{K}. \tag{3.3}$$

4) Mel filterbank energies are obtained by triangular filters spaced on the Mel scale:

$$S[f,m] = \sum_{k=0}^{K-1} B[f,k]\, P[k,m], \quad 1 \le f \le F. \tag{3.4}$$

The Mel frequency mapping for a linear frequency ν (in Hz) is

$$\mathrm{mel}(\nu) = 2595 \,\log_{10}\left(1 + \frac{\nu}{700}\right), \tag{3.5}$$

and the filterbank $\mathbf{B}$ is formed by piecewise-linear triangular filters uniformly spaced on the Mel axis and then mapped back to linear frequency bins.

5) Logarithmic compression reduces dynamic range and approximates loudness perception:

$$\widehat{S}[f,m] = \log\big(S[f,m] + \varepsilon\big). \tag{3.6}$$

6) DCT-II decorrelates the log-Mel energies, yielding MFCCs:

$$c_q[m] = \sum_{f=1}^{F} \widehat{S}[f,m] \cos\left(\frac{\pi q}{F}\left(f - \tfrac{1}{2}\right)\right), \quad 0 \le q < Q, \tag{3.7}$$

where Q is the number of cepstral coefficients retained (commonly $Q \in [12, 20]$). An optional sinusoidal lifter improves energy distribution across coefficients:

$$\widetilde{c}_q[m] = \left(1 + \frac{L}{2} \sin\frac{\pi q}{L}\right) c_q[m], \quad L > 0. \tag{3.8}$$

Temporal dynamics are often captured by regression-based derivatives using a symmetric window of radius R:

$$\Delta c_q[m] = \frac{\sum_{r=1}^{R} r\,\big(c_q[m+r] - c_q[m-r]\big)}{2\sum_{r=1}^{R} r^2}, \quad \Delta\Delta c_q[m] \text{ analogously.} \tag{3.9}$$

Advantages: MFCCs are compact, relatively robust to slowly varying convolutive effects, and widely validated in speech tasks. Figure 3.1 and Figure 3.2 show the MFCCs of a sample in IEMOCAP and RAVDESS. Limitations: information loss due to coarse spectral integration and DCT decorrelation, frame-level stationarity assumptions, and sensitivity to noise and channel mismatch without appropriate normalization or enhancement.

Figure 3.1: The MFCC of a sample in IEMOCAP

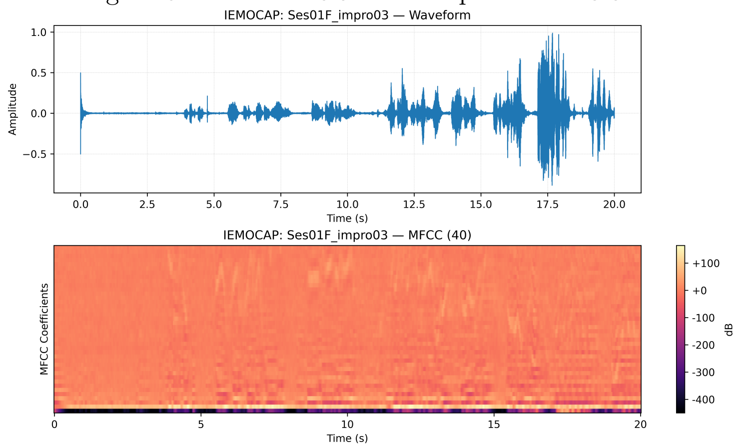
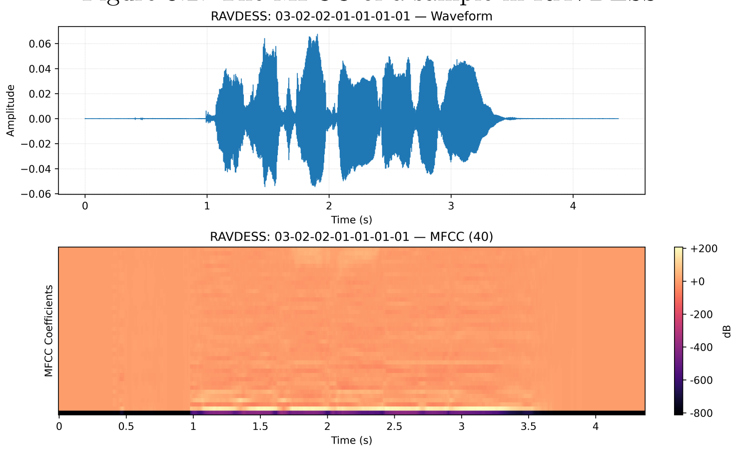


Figure 3.2: The MFCC of a sample in RAVDESS



3.1.2 Mel-spectrogram

The Mel-spectrogram represents short-time spectral energy projected onto a perceptual frequency axis, typically used directly as a 2D input to convolutional or transformer models.

Given the short-time spectrum in (3.2) and power spectrum in (3.3), the Mel-spectrogram is

$$\mathrm{MelSpec}[f,m] = \sum_{k=0}^{K-1} B[f,k]\, P[k,m], \quad 1 \le f \le F. \tag{3.10}$$

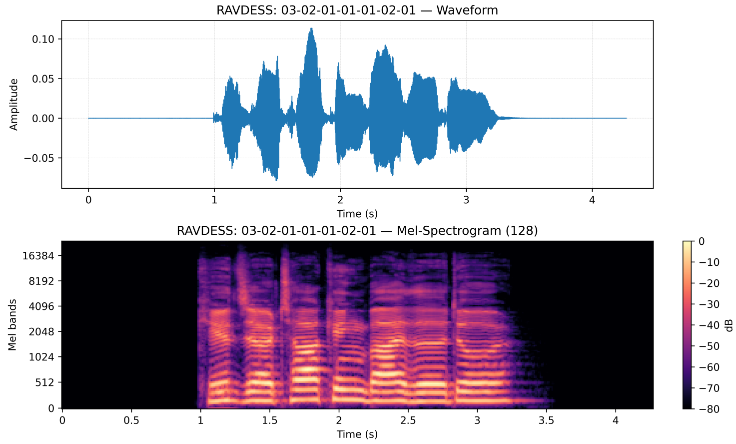
A log or power-law compression is commonly applied to stabilize training and approximate loudness:

$$\mathrm{LogMel}[f,m] = \log\big(\mathrm{MelSpec}[f,m] + \varepsilon\big) \quad \text{or} \quad \mathrm{MelSpec}[f,m]^{\gamma},\ \gamma \in (0,1]. \tag{3.11}$$

Design choices include the sample rate f_s, frame length N, hop size H, window type $w[n]$, number of Mel filters F, lower/upper frequency bounds of the filterbank, and whether to use magnitude or power spectra. Per-utterance or global mean-variance normalization is often applied to reduce channel variability.

Advantages: Mel-spectrograms preserve local time–frequency structure and are highly compatible with CNNs and attention-based encoders; they avoid the information loss introduced by the DCT in MFCC. Figure 3.3 and Figure 3.4 show the Mel-spectrograms of a sample in RAVDESS and IEMOCAP. Limitations: higher dimensionality increases memory and compute cost; the representation remains sensitive to additive noise and reverberation without enhancement or robust training; choices of F, bounds, and f_s affect resolution and cross-dataset transferability.

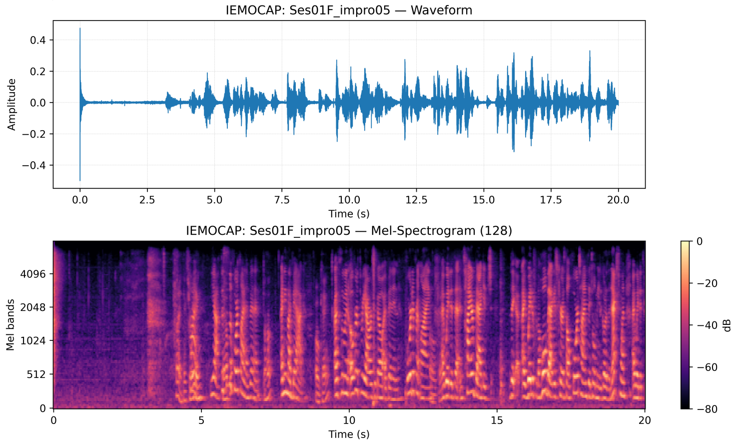
Figure 3.3: The Mel-spectrogram of a sample in RAVDESS



3.2 Model Developments

We consider a label set $\mathcal{C}$ with $|\mathcal{C}| = C$ emotion classes. For utterance-level classification, we denote the per-frame MFCC feature as $\mathbf{z}_m \in \mathbb{R}^Q$ from (3.7) (op-

Figure 3.4: The Mel-spectrogram of a sample in IEMOCAP



tionally concatenated with Δ and $\Delta\Delta$ features from (3.9)). A fixed-dimensional utterance representation is obtained by temporal mean pooling

$$\bar{\mathbf{z}} = \frac{1}{T}\sum_{m=1}^{T} \mathbf{z}_m \in \mathbb{R}^D, \quad (D = Q \text{ or } 3Q), \tag{3.12}$$

optionally augmented with standard deviation pooling. Let $\mathbf{y} \in \{0,1\}^C$ be the one-hot label vector.

3.2.1 Multi-layer Perceptron

We adopt a feed-forward multilayer perceptron (MLP, as shown in Figure 3.5) as a lightweight baseline operating on MFCC-based utterance vectors. Given input $\bar{\mathbf{z}}$ from (3.12), a depth-L MLP computes

$$\mathbf{h}^{(1)} = \sigma\big(\mathbf{W}^{(1)}\bar{\mathbf{z}} + \mathbf{b}^{(1)}\big), \tag{3.13}$$

$$\mathbf{h}^{(\ell)} = \sigma\big(\mathbf{W}^{(\ell)}\mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)}\big), \quad 2 \le \ell \le L-1, \tag{3.14}$$

$$\mathbf{o} = \mathbf{W}^{(L)}\mathbf{h}^{(L-1)} + \mathbf{b}^{(L)}, \tag{3.15}$$

where $\sigma(\cdot)$ is a pointwise nonlinearity (ReLU by default), $\mathbf{W}^{(\ell)}$ and $\mathbf{b}^{(\ell)}$ are trainable parameters. Class probabilities are

$$\mathbf{p} = \operatorname{softmax}(\mathbf{o}), \quad p_c = \frac{\exp(o_c)}{\sum_{c'=1}^{C}\exp(o_{c'})}. \tag{3.16}$$

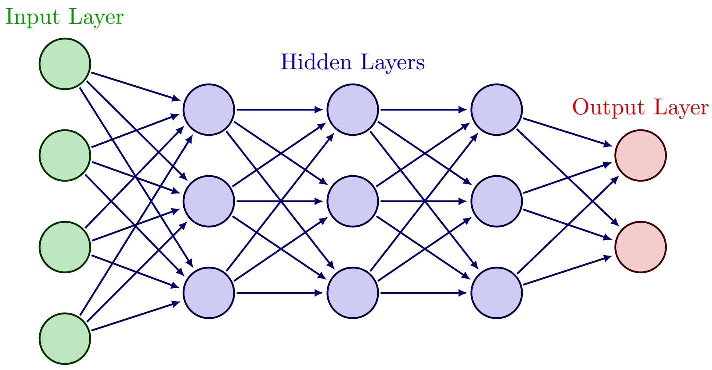
We train with cross-entropy and ℓ_2 regularization:

$$\mathcal{L} = -\sum_{c=1}^{C} y_c \log p_c + \lambda\|\Theta\|_2^2, \tag{3.17}$$

where Θ collects all parameters and λ controls weight decay. Dropout between hidden layers improves generalization.

Role in this thesis: the MLP serves as a parameter-efficient baseline with MFCC inputs, quantifying the incremental benefits of time–frequency 2D models.

Figure 3.5: The architecture of MLP



3.2.2 Shallow CNN

For time–frequency inputs we use the log-Mel representation LogMel $\in \mathbb{R}^{F \times T}$ from (3.11). A shallow 2D CNN treats LogMel as a single-channel image: $\mathbf{X} \in \mathbb{R}^{1 \times F \times T}$. Let a 2D convolution with kernel $\mathbf{W} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k_f \times k_t}$, stride (s_f, s_t), and padding (p_f, p_t) be

$$(\mathbf{W} * \mathbf{X})[c, f, t] = \sum_{c'=1}^{C_{\text{in}}} \sum_{i=0}^{k_f - 1} \sum_{j=0}^{k_t - 1} \mathbf{W}[c, c', i, j]\, \mathbf{X}[c', f + i - p_f,\, t + j - p_t]. \quad (3.18)$$

Each convolutional block applies convolution, batch normalization, ReLU, and pooling:

$$\mathbf{U}_1 = \mathrm{ReLU}\big(\mathrm{BN}(\mathbf{W}_1 * \mathbf{X})\big), \qquad \mathbf{V}_1 = \mathrm{Pool}(\mathbf{U}_1), \quad (3.19)$$

$$\mathbf{U}_2 = \mathrm{ReLU}\big(\mathrm{BN}(\mathbf{W}_2 * \mathbf{V}_1)\big), \qquad \mathbf{V}_2 = \mathrm{Pool}(\mathbf{U}_2). \quad (3.20)$$

We then apply global average pooling over time and frequency and a linear classifier to obtain logits $\mathbf{o}$ followed by (3.16).

Design: we use two convolutional blocks with moderate kernels (e.g., $k_f \in \{3, 5\}$, $k_t \in \{3, 5\}$), stride 1, and max-pooling with small windows to preserve resolution. Regularization includes dropout and additive noise during training to improve robustness.

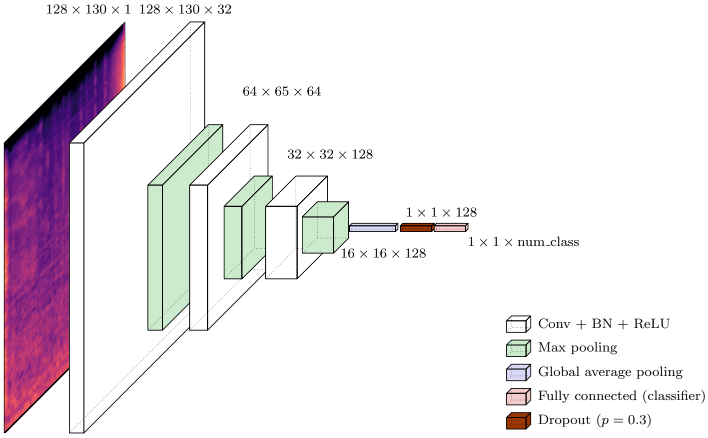
Advantages: shallow CNNs are data-efficient and fast, capturing local spectro-temporal patterns. Limitations: weaker long-range modeling and limited receptive field without deeper stacks or dilations. In this thesis, the shallow CNN consumes log-Mel features to quantify the benefits of local 2D structure over MFCC-based MLPs.

3.2.3 ResNet-18

We adopt ResNet-18 as a deeper convolutional baseline to enhance receptive field and hierarchical feature learning on LogMel. A basic residual block computes

$$\mathbf{y} = \mathcal{F}(\mathbf{x}; \{\mathbf{W}_i\}) + \mathcal{S}(\mathbf{x}), \quad (3.21)$$

Figure 3.6: The architecture of shallow CNN



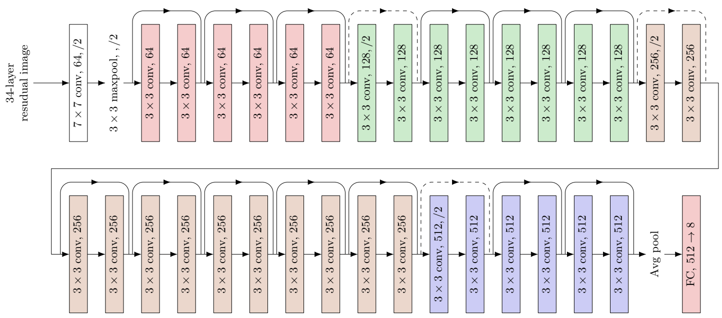
where $\mathcal{F}$ is a stack of Conv→BN→ReLU→Conv→BN, and $\mathcal{S}$ is either identity or a 1×1 projection to match shape when stride/downsampling is used. The network stacks such blocks with increasing channels and occasional stride-2 to downsample along time and frequency. After global average pooling, a linear classifier produces logits $\mathbf{o}$ and probabilities via (3.16).

For spectrogram inputs, the first convolution is adapted to single-channel input ($C_{\text{in}} = 1$) with kernel size 7×7 and stride 2, followed by batch normalization and ReLU. Weight decay and label smoothing may be applied to improve generalization.

Advantages: ResNet-18 balances capacity and efficiency, modeling broader context and invariances. Limitations: increased compute relative to shallow CNNs and potential overfitting on small corpora without strong regularization. In this thesis, ResNet-18 processes LogMel to assess the gains of residual learning.

Figure 3.7 shows the architecture of ResNet-18.

Figure 3.7: The architecture of ResNet-18



3.2.4 Audio Spectrogram Transformer

The Audio Spectrogram Transformer (AST) adapts the Vision Transformer (ViT) to audio by treating the log-Mel spectrogram as a 2D patch sequence. Given LogMel $\in \mathbb{R}^{F\times T}$, we partition it into non-overlapping patches of size $P_f \times P_t$, yielding $N = \lfloor F/P_f \rfloor \cdot \lfloor T/P_t \rfloor$ patches. Each patch is flattened and linearly projected to a token embedding:

$$\mathbf{z}_i = \mathrm{vec}(\mathrm{patch}_i)\,\mathbf{W}_e + \mathbf{b}_e \in \mathbb{R}^d, \quad i = 1, \ldots, N. \tag{3.22}$$

A learnable classification token $\mathbf{z}_{\text{cls}}$ is prepended and positional embeddings $\mathbf{p}_i$ are added. The resulting sequence $\{\mathbf{z}_{\text{cls}}, \mathbf{z}_1, \ldots, \mathbf{z}_N\}$ is processed by a stack of transformer encoder layers, each composed of multi-head self-attention (MHSA) and a positionwise feed-forward network (FFN) with residual connections and layer normalization. For a single head,

$$\mathrm{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}, \tag{3.23}$$

with queries $\mathbf{Q} = \mathbf{Z}\mathbf{W}^Q$, keys $\mathbf{K} = \mathbf{Z}\mathbf{W}^K$, values $\mathbf{V} = \mathbf{Z}\mathbf{W}^V$, and $\mathbf{Z}$ the input token matrix; MHSA concatenates multiple heads. The FFN uses two linear layers and a nonlinearity (e.g., GELU). The classification head reads the [CLS] token to produce logits $\mathbf{o}$ and probabilities via (3.16).

Design: patch sizes (P_f, P_t) balance spectral and temporal resolution; smaller patches increase sequence length and cost but improve detail. Regularization includes dropout and stochastic depth; data augmentation includes time/frequency masking and additive noise.

Advantages: AST captures global dependencies and long-range context beyond CNN receptive fields. Limitations: higher computational cost and sensitivity to data scale; careful regularization and pretraining can be beneficial. In this thesis, AST consumes log-Mel inputs to examine the benefits of attention-based modeling.

Figure 3.8 shows the architecture of AST.

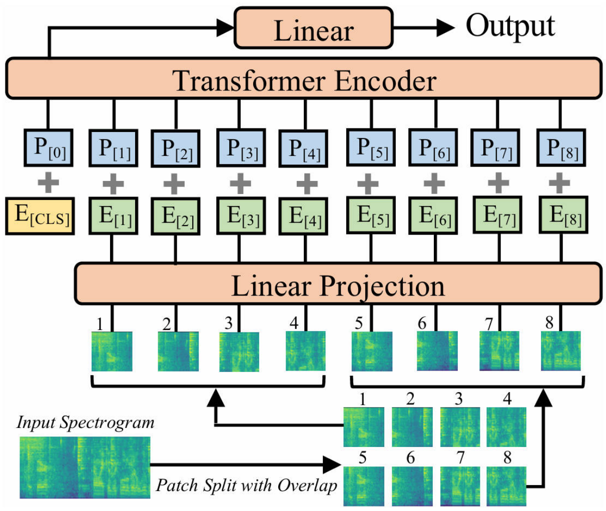
3.3 Data Augmentation

Deep models for SER often overfit clean, laboratory speech and degrade under domain shift caused by background noise, channel mismatch, and reverberation. To improve generalization from clean training conditions to noisy evaluations, we perform on-the-fly additive-noise augmentation during training while keeping the validation and clean test sets unchanged. We follow the unified notation in Section 6: the discrete-time speech signal is $x[n]$ at sample rate f_s, and we denote a noise signal by $v[n]$. All mixing is performed in the time domain prior to feature extraction (MFCC or Log-Mel), ensuring that downstream features consistently reflect the augmented acoustics.

3.3.1 Noise sources and SNR set

We consider two noise families: (i) white Gaussian noise as a controlled baseline; and (ii) environmental recordings sampled from ESC-50 (three representative classes). The target signal-to-noise ratio (SNR) in decibels is uniformly sampled

Figure 3.8: The architecture of AST



from $\mathcal{S} = \{0, 5, 10, 20\}$ dB. When comparing architectures, we additionally use a fixed 20 dB white-noise setting as a reference point.

3.3.2 On-the-fly sampling policy

For each training utterance, with probability $p_{\text{aug}} = 0.7$ we inject one type of noise, otherwise we keep the sample clean. The noise family (white vs. ESC-50 subset) and the SNR are sampled uniformly. Each speech sample is mixed with at most one noise instance.

3.3.3 Time alignment and preprocessing

Given a speech clip of length T samples and a noise recording $v[n]$: if the noise is shorter than T, we tile it by concatenation or loop with a random starting offset; if it is longer, we randomly crop a segment of length T. Both speech and noise are resampled to f_s if needed and centered to zero mean. Let $x_s[n]$ and $x_n[n]$ denote the length-matched speech and noise signals, respectively, for $0 \leq n < T$.

3.3.4 RMS, target SNR, and gain computation

We use the root-mean-square (RMS) to control the noise gain. For a discrete signal $z[n]$ of length T:

$$R(z) = \sqrt{\frac{1}{T} \sum_{n=0}^{T-1} z[n]^2}. \tag{3.24}$$

Let $R_s = R(x_s)$ and $R_n = R(x_n)$. Given a desired SNR in decibels, SNR_{dB}, the linear SNR is $\text{SNR}_{\text{lin}} = 10^{\text{SNR}_{\text{dB}}/20}$. We choose a scalar gain g such that the mixture achieves the target SNR:

$$g = \frac{R_s}{R_n\,\mathrm{SNR}_{\mathrm{lin}}} = \frac{R_s}{R_n \cdot 10^{\mathrm{SNR}_{\mathrm{dB}}/20}}. \tag{3.25}$$

The mixture is

$$x_{\mathrm{mix}}[n] = x_s[n] + g\,x_n[n]. \tag{3.26}$$

By construction, the realized SNR satisfies

$$\mathrm{SNR}_{\mathrm{dB}} = 20\,\log_{10}\!\left(\frac{R(x_s)}{R(g\,x_n)}\right) = 20\,\log_{10}\!\left(\frac{R_s}{g\,R_n}\right). \tag{3.27}$$

3.3.5 Peak limiting and re-normalization

To prevent clipping after mixing, we enforce a peak limit of $-1\,\mathrm{dBFS}$. Let $A_{\mathrm{max}} = 10^{-1/20}$ be the amplitude threshold and $A_{\mathrm{peak}} = \max_n |x_{\mathrm{mix}}[n]|$. If $A_{\mathrm{peak}} > A_{\mathrm{max}}$, we rescale

$$\widetilde{x}_{\mathrm{mix}}[n] = x_{\mathrm{mix}}[n]\,\frac{A_{\mathrm{max}}}{A_{\mathrm{peak}}}, \tag{3.28}$$

and otherwise leave the mixture unchanged. A final mean-variance normalization per utterance can be applied before feature extraction for numerical stability.

3.3.6 Integration with the training pipeline

The augmentation is performed online per mini-batch before computing features: we form either MFCCs or Log-Mel spectrograms from $\widetilde{x}_{\mathrm{mix}}[n]$ or $x_s[n]$ depending on whether augmentation was applied. This preserves consistency of notation and ensures identical downstream preprocessing. The same policy is shared by all architectures (MLP, shallow CNN, ResNet-18, AST) to enable fair comparison.

3.3.7 Evaluation protocol

We adopt RAVDESS (speech actors 01–24) as the primary dataset with the existing data split strategy. Augmentation is applied only to the training split. We evaluate on: (i) the original clean test set to verify no degradation on matched conditions; and (ii) noisy test sets generated by mixing the clean test audio with held-out noise at SNRs in $\mathcal{S}$. We report standard metrics for categorical SER (e.g., accuracy and unweighted average recall) and analyze performance as a function of SNR to identify robustness trends and breakpoints. A fixed 20 dB white-noise protocol is additionally used as a reference to compare architectural robustness, with particular attention to transformer-based AST versus conventional CNN/MLP baselines.

CHAPTER 4

Experiments

This chapter presents a comprehensive experimental study on SER under clean and noisy conditions. We evaluate multiple architectures and noise augmentation strategies on two datasets, adhering to the unified notation and preprocessing defined in Chapter 3. Unless otherwise specified, audio is sampled at f_s, the time-domain signal is $x[n]$, noise is $v[n]$, and on-the-fly mixtures during training follow $x_{\text{mix}}[n] = x_s[n] + g\, x_n[n]$ with gain g computed from target SNR as in Chapter 3.

4.1 Common Settings

4.1.1 Datasets and splits

We use RAVDESS (speech actors 01–24) and IEMOCAP. Data splits follow the established protocol for each dataset. Validation and clean test sets are never augmented; only the training split is subjected to on-the-fly noise mixing when enabled. The batch size of MLP, shallow CNN, and Resnet18 is 128, and 64 for AST. All models are trained using Adam [17] optimizer with a initial learning rate of 0.001. The training is terminated when the validation loss does not improve for 10 epochs.

4.1.2 Models

We compare a convolutional baseline (ResNet-18 on Log-Mel spectrograms) and a transformer-based model (AST Base on filterbank features), alongside lighter baselines where applicable (MLP and shallow CNN) to contextualize complexity–performance trade-offs.

4.1.3 Training and evaluation

Training uses mini-batch stochastic optimization with standard regularization. Metrics on the validation set include accuracy, precision, recall, and macro-F1. Confusion matrices are reported to analyze class-wise separability. Robustness is further assessed on noisy test sets created by offline mixing of $x[n]$ with held-out noise at specified SNRs $\in \{0, 5, 10, 20\}$ dB. All experiments fix a base random seed 42 unless otherwise stated.

4.1.4 Noise augmentation policy

When enabled, augmentation follows Chapter 3: with probability p_{aug}, we sample a noise type (white or ESC-50 subsets), sample SNR from $\{0, 5, 10, 20\}$ dB, time-align noise to the utterance, compute gain g via RMS matching to satisfy the target SNR, form $x_{\text{mix}}[n]$, apply peak limiting at -1 dBFS, and optionally mean–variance normalize prior to feature extraction.

4.2 Experiment A: Clean Baselines

Objective. Establish upper bounds under matched clean conditions to isolate modeling capacity from noise robustness. These baselines indicate how well each architecture fits clean speech and provide a reference point for subsequent robustness interventions.

Setup. All training, validation, and test audio remain unaltered. Feature pipelines follow Chapter 3 (MFCC and log-mel) as required by each model. Optimization hyperparameters are fixed across runs for comparability. No augmentation, denoising, or calibration is applied. Training–validation trajectories are examined to gauge overfitting and learning dynamics.

Evaluation. We report accuracy, precision, recall, and macro-F1, with confusion matrices to visualize class confusability. Macro-F1 is emphasized due to class imbalance, ensuring that performance is not dominated by frequent emotions. Where informative, we comment on calibration and thresholding effects.

Results and analysis. Results of Experiment A on RAVDESS and IEMOCAP are shown in Table 4.1 and Table 4.2 respectively. Figure 4.1 provides a more intuitive view of the results. On RAVDESS, AST Base achieves the best performance (Acc 0.718, Macro-F1 0.714), closely followed by ResNet-18 (Acc 0.687). MLP offers a moderate reference (Acc 0.451), while the shallow CNN underfits substantially (Acc 0.208), suggesting inadequate receptive-field depth for clean spectral patterns. On IEMOCAP, AST again leads in accuracy (0.615) but exhibits a precision–recall asymmetry (Precision 0.682 vs. Recall 0.413), indicative of conservative predictions that miss minority or ambiguous classes. ResNet-18 is competitive yet shows greater variability across sessions, and MLP/CNN display sensitivity to the dataset's heterogeneity and longer utterances. Confusions concentrate among emotions with overlapping prosody (e.g., neutral–sad, angry–excited), consistent with prior findings. Training–validation comparisons suggest mild overfitting for deeper models on RAVDESS and larger variance on IEMOCAP, pointing to limited per-class data. These ceilings highlight headroom primarily in recall on IEMOCAP and in narrowing the AST–ResNet gap on RAVDESS.

Table 4.1: Results of Experiment A on RAVDESS

Model	Accuracy	Precision	Recall	Macro-F1
MLP	0.451	0.412	0.431	0.402
Shallow CNN	0.208	0.141	0.192	0.127
ResNet-18	0.687	0.694	0.697	0.682
AST Base	0.718	0.735	0.719	0.714

Table 4.2: Results of Experiment A on IEMOCAP

Model	Accuracy	Precision	Recall	Macro-F1
MLP	0.583	0.498	0.342	0.340
Shallow CNN	0.583	0.546	0.449	0.415
ResNet-18	0.570	0.388	0.475	0.406
AST Base	0.615	0.682	0.413	0.448

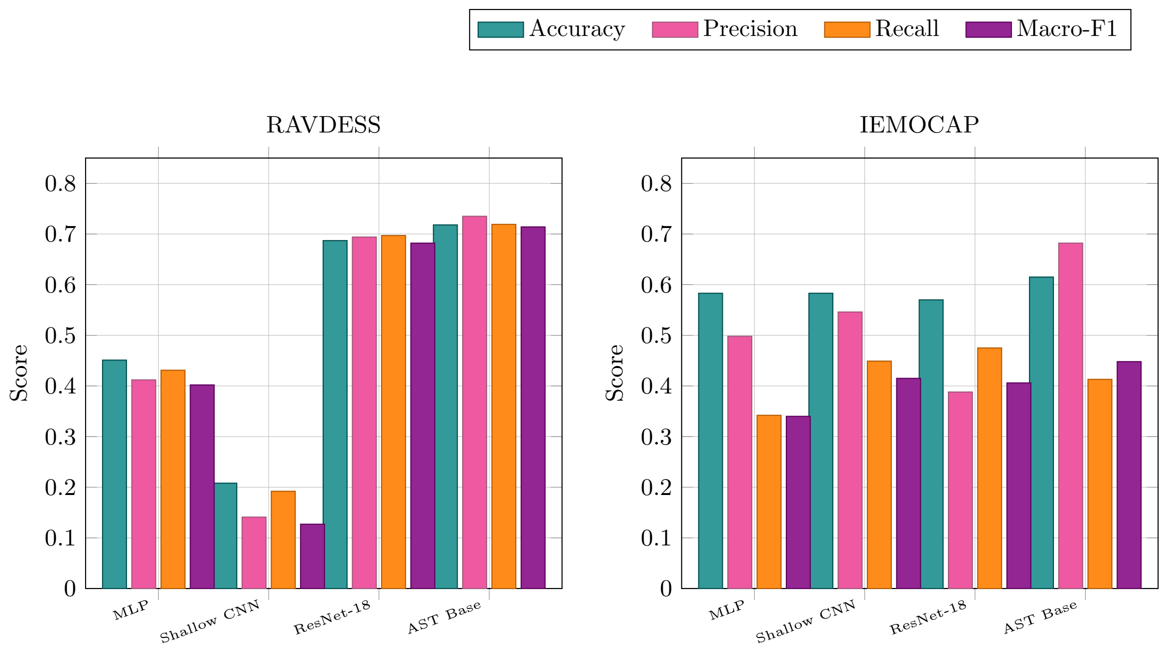


Figure 4.1: Experiment A: Grouped bar charts of four models across four metrics on RAVDESS and IEMOCAP.

4.3 Experiment B: White-Noise Training at 20 dB

Objective. Test whether injecting mild additive white noise during training (SNR = 20 dB) enhances robustness while preserving accuracy on clean evaluation. We also examine how this augmentation shifts the precision–recall balance and class confusability relative to Experiment A.

Setup. During training, white noise at SNR = 20 dB is mixed with probability p_{aug} using RMS-based gain control (Chapter 3). Validation and the primary test set remain clean to isolate changes in clean-set generalization. Hyperparameters and feature configurations are held fixed across experiments for fair comparison.

Evaluation. We report accuracy, precision, recall, and macro-F1, emphasizing macro-F1 to account for class imbalance. Results are compared directly against Experiment A to quantify any trade-offs introduced by augmentation.

Results and analysis. Table 4.3 shows the results of Experiment B on RAVDESS and IEMOCAP. On RAVDESS, clean-set performance is essentially preserved for AST Base (Acc 0.718, Macro-F1 0.713 vs. 0.718/0.714 in Exp A), indicating that 20 dB noise does not harm its clean discriminability. ResNet-18 exhibits a small decline (Acc 0.677, Macro-F1 0.674 vs. 0.687/0.682), consistent with a mild regularization effect that slightly shifts decision boundaries without substantial loss. On IEMOCAP, effects are model-dependent. ResNet-18 shows higher precision (0.491 vs. 0.388) but lower recall (0.378 vs. 0.475) and macro-F1 (0.315 vs. 0.406), suggesting a more conservative operating regime that reduces false positives at the expense of coverage, potentially reflecting session and speaker variability. In contrast, AST Base improves recall (0.471 vs. 0.413) and macro-F1 (0.472 vs. 0.448) while experiencing lower accuracy (0.583 vs. 0.615) and precision (0.565 vs. 0.682). This pattern indicates better balance across classes—even if top-1 correctness declines—consistent with augmentation helping the transformer capture noise-invariant cues on a heterogeneous corpus. Overall, training with 20 dB white noise acts as a gentle regularizer: it preserves clean-set performance on the simpler RAVDESS distribution and yields dataset-dependent precision–recall shifts on IEMOCAP, with net gains in macro-F1 for AST and modest degradations for ResNet-18.

Table 4.3: Results of Experiment B on two datasets

	RAVDESS				IEMOCAP			
Model	Acc.	Prec.	Recall	Macro-F1	Acc.	Prec.	Recall	Macro-F1
ResNet-18	0.677	0.674	0.680	0.674	0.551	0.491	0.378	0.315
AST Base	0.718	0.717	0.716	0.713	0.583	0.565	0.4713	0.472

4.4 Experiment C: ESC-50 Multi-Category, Multi-SNR

Objective. Evaluate whether multi-category, multi-SNR environmental noise augmentation (ESC-50; natural soundscapes and human non-speech, SNR $\in$

$\{0, 5, 10, 20\}$ dB) builds noise-invariant representations while preserving or improving clean-set performance. We compare architectures and datasets to quantify precision–recall shifts under broader nuisance variability.

Setup. During training, a noise category and SNR are sampled uniformly, and mixed via RMS-controlled gain with peak limiting at -1 dBFS (Chapter 3). Validation and the primary test sets remain clean; noisy test sets are additionally constructed and stratified by category and SNR. Feature pipelines and hyperparameters are kept fixed for fair comparison across experiments.

Evaluation. We report accuracy, precision, recall, and macro-F1, emphasizing macro-F1 to mitigate class-imbalance effects. Results are contrasted with Experiments A/B to isolate the impact of realistic, heterogeneous augmentation from white-noise regularization.

Results and analysis. Table 4.4 shows the results of Experiment C on RAVDESS and IEMOCAP. On RAVDESS, AST Base improves over the clean baseline (Acc 0.739 vs. 0.718; Macro-F1 0.734 vs. 0.714), indicating that diverse ESC-50 perturbations enhance generalization without harming clean discriminability. In contrast, ResNet-18 degrades notably (Acc 0.548; Macro-F1 0.527 vs. 0.687/0.682 in Exp A), suggesting sensitivity to the broader noise distribution and possible misalignment between convolutional inductive biases and heterogeneous background cues. On IEMOCAP, AST Base again benefits: recall rises (0.488 vs. 0.413) and macro-F1 increases (0.515 vs. 0.448), with a slight accuracy gain (0.625 vs. 0.615) and modest precision reduction (0.645 vs. 0.682). This pattern reflects improved coverage of minority or ambiguous classes, consistent with more balanced decision boundaries under augmentation. By contrast, ResNet-18 exhibits a pronounced precision–recall trade-off (precision 0.519↑, recall 0.327↓; macro-F1 0.323↓ from 0.406), indicating a conservative operating regime that reduces false positives at the expense of sensitivity. Overall, multi-SNR, multi-category augmentation appears especially effective for the transformer (AST) architecture, yielding higher ceilings on the simpler RAVDESS distribution and recall-oriented gains on the heterogeneous IEMOCAP corpus, while the CNN (ResNet-18) shows vulnerability to over-regularization or distributional mismatch.

Table 4.4: Results of Experiment C on two datasets

	RAVDESS				IEMOCAP			
Model	Acc.	Prec.	Recall	Macro-F1	Acc.	Prec.	Recall	Macro-F1
ResNet-18	0.548	0.541	0.546	0.527	0.554	0.519	0.327	0.323
AST Base	0.739	0.746	0.745	0.734	0.625	0.645	0.488	0.515

4.5 Experiment D: SNR Ablation (Natural Soundscapes)

Objective. Evaluate whether multi-category, multi-SNR environmental noise augmentation (ESC-50; natural soundscapes and human non-speech, SNR $\in$ $\{0, 5, 10, 20\}$ dB) builds noise-invariant representations while preserving or improving clean-set discriminability. The experiment compares architectures and datasets to quantify shifts in accuracy, precision–recall balance, and macro-

F1 under heterogeneous nuisance variability, and to assess whether the learned decision boundaries become more class-balanced than those obtained in Experiments A/B.

Setup. During training, a noise category and an SNR are sampled uniformly and mixed through RMS-controlled gain with peak limiting at -1 dBFS (Chapter 3). Validation and the primary test sets remain clean to isolate changes in clean-set generalization; additional noisy test sets are constructed and stratified by category and SNR for robustness evaluation. Feature pipelines and optimization hyperparameters are held fixed across experiments to ensure comparability.

Evaluation. We report accuracy, precision, recall, and macro-F1. Macro-F1 is emphasized to mitigate class-imbalance artifacts and to reflect model behavior on minority and ambiguous emotions. Qualitative analyses use confusion matrices and heatmaps (model $\times$ noise $\times$ SNR) to visualize boundary shifts and category-specific confusability. Aggregate results are compared with Experiments A (clean baselines) and B (white-noise at 20 dB) to contextualize the impact of realistic environmental noise.

Results and analysis. Table 4.5 and Table 4.6 show the results of Experiment D on RAVDESS and IEMOCAP respectively. Line charts 4.2 provide a more intuitive view of the results. On RAVDESS, AST Base surpasses the clean baseline, achieving Acc 0.739 and Macro-F1 0.734 (vs. 0.718/0.714 in Experiment A). Precision and recall remain well balanced (0.746/0.745), indicating that broad ESC-50 perturbations help AST maintain clean discriminability while enhancing coverage across classes. In contrast, ResNet-18 shows a marked decrease to Acc 0.548 and Macro-F1 0.527 (vs. 0.687/0.682), with precision and recall (0.541/0.546) remaining roughly symmetric but at a lower operating point. This gap suggests that the transformer benefits from diverse temporal–spectral cues introduced by realistic noise, whereas the convolutional model is more sensitive to the heterogeneity and may over-regularize away discriminative fine structure on the simpler RAVDESS distribution.

On IEMOCAP, AST Base again improves relative to Experiment A: Acc increases to 0.625 (from 0.615), recall rises to 0.488 (from 0.413), and Macro-F1 reaches 0.515 (from 0.448), while precision decreases moderately to 0.645 (from 0.682). The pattern reflects a more balanced decision surface that captures minority or ambiguous emotions without substantial loss of overall correctness. ResNet-18, however, shifts to a conservative regime: precision increases to 0.519 (from 0.388), but recall drops to 0.327 (from 0.475), bringing Macro-F1 down to 0.323 (from 0.406). This precision–recall asymmetry indicates tighter boundaries that reduce false positives at the cost of sensitivity, a behavior consistent with session and speaker variability interacting with heterogeneous noise.

Relative to white-noise training (Experiment B), multi-category augmentation yields clearer benefits for AST. On RAVDESS, AST's Macro-F1 increases further (0.734 vs. 0.713), and on IEMOCAP, Macro-F1 improves as well (0.515 vs. 0.472), with gains primarily driven by higher recall. For ResNet-18, the multi-category setting underperforms white-noise on RAVDESS (Macro-F1 0.527 vs. 0.674), while on IEMOCAP its Macro-F1 is similar in magnitude (0.323 vs. 0.315) but exhibits a stronger precision bias. Taken together, these results show that multi-SNR, multi-category augmentation is especially effective for the transformer architecture, which maintains or raises clean-set performance on RAVDESS and improves recall and macro-F1 on the more heterogeneous IEMOCAP corpus. The CNN, by contrast, displays model-dependent trade-offs and a greater susceptibility to the diversity of environmental noise, as reflected by reduced macro-F1 and altered calibration on clean evaluation.

Table 4.5: Experiment D (RAVDESS): SNR ablation with natural soundscapes for ResNet-18 and AST. Metrics reported are accuracy (Acc.) and macro-F1 per SNR.

SNR (dB)	ResNet-18 Acc.	ResNet-18 Macro-F1	AST Acc.	AST Macro-F1
0	0.625	0.619	0.718	0.714
5	0.681	0.671	0.736	0.716
10	0.673	0.667	0.739	0.718
20	0.680	0.670	0.746	0.731

Table 4.6: Experiment D (IEMOCAP): SNR ablation with natural soundscapes for ResNet-18 and AST. Metrics reported are accuracy (Acc.) and macro-F1 per SNR.

SNR (dB)	ResNet-18 Acc.	ResNet-18 Macro-F1	AST Acc.	AST Macro-F1
0	0.570	0.406	0.615	0.448
5	0.577	0.366	0.624	0.460
10	0.596	0.370	0.642	0.526
20	0.609	0.351	0.636	0.457

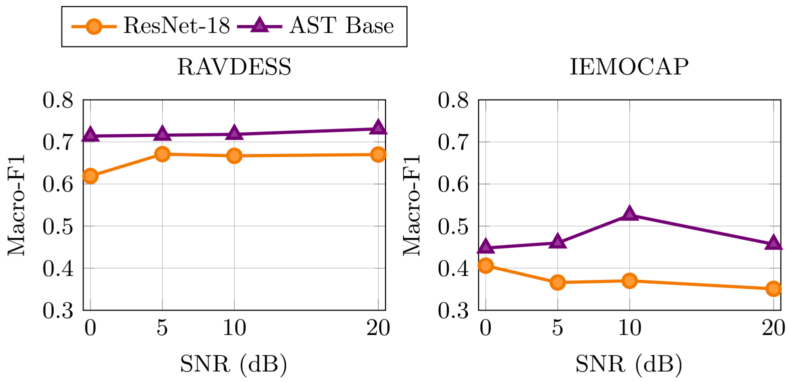


Figure 4.2: Experiment D: Macro-F1 vs. SNR with natural soundscapes for ResNet-18 and AST Base.

CHAPTER 5

Discussion

1. Architecture–augmentation interaction. Across experiments, the transformer (AST Base) consistently benefits from noise augmentation, whereas the CNN (ResNet-18) is more fragile to heterogeneous backgrounds. With ESC-50 multi-category, multi-SNR noise, AST improves over clean baselines on both datasets (RAVDESS Macro-F1: 0.734 vs. 0.714; IEMOCAP Macro-F1: 0.515 vs. 0.448), while ResNet-18 drops on RAVDESS (Macro-F1: 0.527 vs. 0.682). Under white-noise at 20 dB, AST preserves clean performance on RAVDESS and raises Macro-F1 on IEMOCAP, but ResNet-18 exhibits small degradations. These patterns suggest that attention-based models better absorb diverse temporal–spectral perturbations, whereas convolutional inductive biases may over-regularize discriminative fine structure under broad noise.

2. Precision–recall balance under dataset heterogeneity. IEMOCAP exhibits stronger class imbalance and session/speaker variability than RAVDESS, and augmentation shifts the operating point differently per model. With ESC-50, AST trades a modest precision decrease (0.682→0.645) for a marked recall gain (0.413→0.488), raising Macro-F1 to 0.515; this indicates improved coverage of minority or ambiguous emotions. ResNet-18 shows the opposite shift (precision 0.388→0.519; recall 0.475→0.327), lowering Macro-F1 to 0.323 and implying a conservative regime that reduces false positives at the expense of sensitivity. On RAVDESS, both models keep precision–recall relatively balanced, highlighting that corpus complexity largely dictates the calibration trade-off.

3. SNR sensitivity and non-monotonic trends. Under natural soundscapes (Experiment D), AST on RAVDESS shows a gentle improvement with SNR (Macro-F1 0.714→0.731 from 0 to 20 dB), while on IEMOCAP it peaks at 10 dB (0.526) and slightly declines at 20 dB (0.457), revealing a sweet spot where augmentation regularizes without oversuppressing informative cues. ResNet-18 varies little on RAVDESS (0.619–0.671) but

degrades with increasing SNR on IEMOCAP (0.406 at 0 dB to 0.351 at 20 dB), consistent with increased decision conservativeness in challenging conditions. Together, these results indicate that robustness depends jointly on the SNR regime and dataset variability, with mid-SNR mixes often providing the best balance between invariance and class separability for AST.

CHAPTER 6

Conclusion

This dissertation set out to investigate and improve the performance of Speech Emotion Recognition systems, with a specific focus on enhancing model robustness in noisy environments. Through a systematic series of experiments, we conducted a comparative analysis of multiple feature representations (MFCCs and Mel-spectrograms) and four distinct deep learning architectures: MLP, a shallow CNN, ResNet-18, and the Audio Spectrogram Transformer (AST). The central contribution of this work lies in the extensive evaluation of on-the-fly noise augmentation as a mechanism for improving generalization, using both the RAVDESS and IEMOCAP datasets as testbeds. The main findings and their implications are summarized below.

The experimental results provide conclusive evidence for the architectural superiority of the Audio Spectrogram Transformer. Across all conditions, AST consistently outperformed the other models, achieving the highest accuracy and macro-F1 scores on both clean and noise-augmented test sets. This highlights the effectiveness of its self-attention mechanism in capturing global spectro-temporal dependencies that are crucial for emotion discrimination. Furthermore, on-the-fly noise augmentation proved to be a powerful and effective regularizer. For the AST model, training with a diverse set of natural soundscapes not only fortified its performance against noise but also led to improved results on the original clean data, suggesting that the perturbations encouraged the model to learn more generalizable and salient affective features.

A key insight from this work is the strong interaction between model architecture and the augmentation strategy. The benefits of augmentation were most pronounced for the AST model, which successfully absorbed the acoustic variability to enhance its representations. In contrast, the ResNet-18 architecture, despite its depth, showed fragility and performance degradation when exposed to the same heterogeneous noise, indicating that its convolutional inductive biases may be less suited for disentangling emotional content from complex background sounds. This dissertation also confirmed that performance is highly dependent on dataset characteristics; on the more challenging and spontaneous IEMOCAP dataset, augmentation prompted the AST model to achieve a better precision-recall balance, improving its ability to recognize ambiguous or minority-class emotions.

While this study provides a clear direction for robust SER, several limitations present avenues for future research. The noise sources, though representative, were limited to white noise and a subset of the ESC-50 dataset; future work could incorporate a wider variety of real-world noises, such as babble, music, and channel reverberation. Additionally, while AST proved effective, its performance could be further enhanced by leveraging self-supervised pre-training on large-scale unlabeled datasets, following frameworks like wav2vec 2.0. Another promising direction is the extension of this framework to multimodal SER, integrating textual or visual information, which is particularly relevant for conversational datasets like MELD. Finally, future investigations could explore model interpretability to better understand how attention mechanisms identify and leverage emotionally salient regions within a spectrogram. In summary, the findings of this thesis provide compelling evidence for the adoption of transformer-based architectures combined with robust data augmentation for building the next generation of effective and reliable Speech Emotion Recognition systems.

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